



## **Arkansas Game and Fish Commission Aerial Waterfowl Survey Report**

### **December 6-9, 2021**

Arkansas Game and Fish Commission staff Jason Carbaugh, Jason Jackson, Cameron Tatom and Alex Zachary conducted the December 2021 aerial waterfowl surveys in the Mississippi Alluvial Valley (Delta), Arkansas River Valley and southwest Arkansas. Biologists conducting transect-based surveys in the Delta estimated 566,456 total ducks, 326,064 of which were mallards (Table 1) and 13,434 ducks in the Arkansas River Valley, including 6,719 mallards. Biologists performing cruise surveys in southwest Arkansas reported an estimated 54,031 ducks with 10,109 mallards (Table 2). Arctic goose population estimates totaled 902,848 light (lesser snow and Ross's) geese and 381,067 greater white-fronted geese in the Delta.

The Delta mallard population estimate was roughly half the 2009-2021 long-term December average of over 601,000 and the second-lowest December estimate since formal surveys began in 2009. Even so, this December's mallard estimate is almost 30 percent greater than the record low December mallard count last year. Total duck population estimates also were about half the long-term average of over 1.13 million (Figure 1). On average, mallards account for about 53 percent of all ducks during December surveys; mallards accounted for 58 percent of all ducks in the Delta during this survey. The highest mallard and total duck estimates were in the Lower White and Cache survey zones, and duck density maps reflect these findings. Even so, mallard "hot spots" were rare (Figures 3 and 4).

Arkansas River Valley mallard and total duck estimates both were below average (Figure 5), with the highest mallard and total duck estimates in the East Dardanelle Reservoir and West Dardanelle Reservoir survey zones. Concentration maps highlight some of these areas but generally show a lack of duck concentration areas (Figures 6 and 7). Common for this time of year in this region, concentration areas on maps should not be interpreted as having high absolute numbers of ducks. The southwest Arkansas cruise survey indicated slightly more ducks than usual for December but fewer than the typical number of mallards.

Much like December 2020, dry conditions continue to play a major role in duck distribution and abundance. Habitat availability remains low as a result of almost no runoff, limited overbank flooding and low amounts of flooded agricultural fields. Habitat is almost exclusively limited to actively managed sites. This typically is the case during November, but signs of improvement often appear by now, or at least the hope of rains and the additional habitat that comes with it. This does not appear to be the situation when evaluating current conditions and short- and long-range weather forecasts.

Private- and public land managers providing managed habitat in areas with those capabilities are playing a major role in supporting current duck populations. Permanent water bodies such as Dardanelle Reservoir appear to be supporting most of the ducks in the Arkansas River Valley. Continuing to provide staggered flooding to make new resources available throughout the wintering period will be critically important until major rains occur and unmanaged flooding increases on the landscape. Carefully managed hunting pressure also is especially important in years like this with expected high proportions of adult ducks and limited habitat. Rain brings mallards to Arkansas, so the best we can do at this point is hope for rain.

Staff will conduct the annual midwinter waterfowl survey the week of Jan. 3.

Table 1. Waterfowl abundance estimates in Arkansas during the late November (Nov), mid-December (Dec), early-January Midwinter Survey (MWS) and late-January (Jan) aerial waterfowl survey periods, 2009-2021, in the Mississippi Alluvial Valley (MAV) using stratified random sampling of transects.

|               |        |             | Survey Zone                        |             |                                |           |                        |         |             |                                |                         |                      |             |           |
|---------------|--------|-------------|------------------------------------|-------------|--------------------------------|-----------|------------------------|---------|-------------|--------------------------------|-------------------------|----------------------|-------------|-----------|
|               |        |             | Bayou Bartholomew -<br>Bayou Boeuf | Bayou Macon | Bayou Meto -<br>Lower Arkansas | Big Creek | Black - Upper<br>White | Cache   | L' Anguille | Lower White -<br>Bayou Des Arc | Little River<br>Ditches | Lower St.<br>Francis | Lower White | MAV Total |
| Survey Period | Nov-09 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 124,065   |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 794,405   |
|               | Dec-09 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 648,955   |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 2,046,969 |
|               | MWS-10 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 2,309,453 |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 2,887,810 |
|               | Jan-10 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 2,063,243 |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 3,153,410 |
|               | Nov-10 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 180,198   |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 1,133,126 |
|               | Dec-10 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 1,247,697 |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 1,860,894 |
|               | MWS-11 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 671,982   |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 1,192,518 |
|               | Jan-11 | Mallards    |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 1,311,245 |
|               |        | Total Ducks |                                    |             |                                |           |                        |         |             |                                |                         |                      |             | 1,786,677 |
|               | Nov-11 | Mallards    | 4,750                              | -           | 15,717                         | 66        | 9,968                  | 47,902  | 7,577       | 10,896                         | 2,432                   | 36                   | 32,736      | 132,080   |
|               |        | Total Ducks | 52,662                             | 19,346      | 174,725                        | 1,367     | 32,914                 | 77,686  | 36,010      | 78,700                         | 40,038                  | 61                   | 114,332     | 627,841   |
|               | Dec-11 | Mallards    | 39,569                             | 2,136       | 90,328                         | 10,161    | 73,576                 | 226,861 | 48,173      | 206,485                        | 367,290                 | 122,032              | 283,418     | 1,470,029 |
|               |        | Total Ducks | 135,903                            | 14,267      | 298,196                        | 32,799    | 171,366                | 306,191 | 94,423      | 360,232                        | 417,990                 | 247,685              | 339,894     | 2,418,946 |
|               | MWS-12 | Mallards    | 7,956                              | 989         | 110,141                        | 87,360    | 35,244                 | 318,991 | 51,493      | 43,618                         | 51,721                  | 8,604                | 37,862      | 753,979   |
|               |        | Total Ducks | 29,124                             | 2,318       | 161,830                        | 161,081   | 51,447                 | 368,370 | 89,139      | 60,802                         | 75,241                  | 51,660               | 65,861      | 1,116,873 |
|               | Jan-12 | Mallards    | 22,365                             | 5,917       | 48,569                         | 82,272    | 47,069                 | 102,400 | 38,682      | 232,214                        | 80,546                  | 11,193               | 82,291      | 753,518   |
|               |        | Total Ducks | 47,985                             | 17,165      | 87,045                         | 114,331   | 128,018                | 162,763 | 105,318     | 321,724                        | 86,482                  | 70,673               | 122,334     | 1,263,838 |
|               | Nov-12 | Mallards    | 2,543                              | 7,176       | 44,732                         | 5,298     | 50,797                 | 112,327 | 97,712      | 14,306                         | 19,136                  | 36,967               | 51,127      | 442,121   |
|               |        | Total Ducks | 11,037                             | 38,220      | 95,784                         | 34,352    | 79,726                 | 171,744 | 164,874     | 68,621                         | 25,852                  | 66,825               | 75,764      | 832,799   |
|               | Dec-12 | Mallards    | 37,887                             | 11,126      | 40,660                         | 4,525     | 157,624                | 54,417  | 45,467      | 8,517                          | 29,542                  | 8,993                | 17,448      | 416,206   |
|               |        | Total Ducks | 121,538                            | 22,648      | 70,813                         | 18,267    | 233,838                | 81,262  | 95,628      | 30,981                         | 35,021                  | 45,649               | 31,270      | 786,915   |
|               | MWS-13 | Mallards    | 30,438                             | 12,508      | 75,690                         | 16,112    | 48,272                 | 57,409  | 32,133      | 20,437                         | 48,267                  | 4,633                | 105,865     | 451,764   |
|               |        | Total Ducks | 54,951                             | 19,145      | 120,222                        | 22,876    | 60,929                 | 84,871  | 68,389      | 27,503                         | 56,231                  | 7,511                | 142,842     | 665,470   |
|               | Jan-13 | Mallards    | 28,836                             | 8,921       | 90,090                         | 36,204    | 93,035                 | 62,369  | 26,058      | 7,344                          | 3,511                   | 93,337               | 27,036      | 476,741   |
|               |        | Total Ducks | 128,058                            | 48,672      | 127,548                        | 48,364    | 138,314                | 103,878 | 52,116      | 9,588                          | 3,665                   | 145,229              | 32,483      | 837,915   |
|               | Nov-13 | Mallards    | 13,582                             | 2,841       | 24,371                         | 2,900     | 25,948                 | 66,501  | 54,163      | -                              | 13,242                  | 1,445                | 39,840      | 244,833   |
|               |        | Total Ducks | 200,157                            | 38,409      | 107,960                        | 18,100    | 148,225                | 111,257 | 99,517      | 49,598                         | 46,545                  | 4,206                | 114,572     | 938,546   |
|               | Dec-13 | Mallards    | 73,158                             | 20,062      | 71,142                         | 7,904     | 72,485                 | 25,429  | 63,845      | 54,023                         | 37,107                  | 27,422               | 22,806      | 475,383   |
|               |        | Total Ducks | 154,707                            | 31,980      | 145,453                        | 26,009    | 98,951                 | 36,088  | 122,202     | 77,353                         | 47,533                  | 33,835               | 60,612      | 834,723   |
|               | MWS-14 | Mallards    | 104,455                            | 33,520      | 164,150                        | 3,070     | 66,080                 | 216,061 | 934         | 56,508                         | 25,124                  | 13,835               | 123,399     | 807,136   |
|               |        | Total Ducks | 114,764                            | 44,313      | 182,263                        | 3,070     | 75,082                 | 247,069 | 1,196       | 80,835                         | 25,124                  | 17,143               | 136,817     | 927,676   |
|               | Nov-14 | Mallards    | 9,409                              | 17,100      | 136,741                        | 22,901    | 34,196                 | 19,077  | 3,454       | 22,216                         | 128,948                 | 69,511               | 84,007      | 547,560   |
|               |        | Total Ducks | 83,914                             | 51,660      | 234,759                        | 80,425    | 70,814                 | 29,520  | 12,382      | 45,023                         | 171,835                 | 80,469               | 132,448     | 993,249   |
|               | Dec-14 | Mallards    | 81,653                             | 48,048      | 53,377                         | 7,836     | 159,637                | 12,105  | 36,370      | 8,308                          | 23,966                  | 16,198               | 172,746     | 620,244   |
|               |        | Total Ducks | 107,261                            | 50,700      | 168,894                        | 12,430    | 212,520                | 18,005  | 72,920      | 15,300                         | 24,196                  | 46,082               | 251,119     | 979,427   |
|               | MWS-15 | Mallards    | 113,960                            | 29,818      | 162,687                        | 99,270    | 110,723                | 25,064  | 31,083      | 10,033                         | 8,855                   | 162,042              | 172,026     | 925,561   |
|               |        | Total Ducks | 130,296                            | 30,988      | 188,203                        | 106,124   | 148,309                | 39,287  | 55,675      | 18,601                         | 8,855                   | 321,514              | 180,142     | 1,227,994 |
|               | Nov-15 | Mallards    | 3,599                              | 43,200      | 17,915                         | 19,253    | 15,382                 | 46,418  | 7,625       | 15,597                         | 9,093                   | 40,889               | 42,941      | 261,912   |
|               |        | Total Ducks | 203,640                            | 120,492     | 126,942                        | 25,333    | 49,581                 | 149,017 | 18,051      | 22,088                         | 14,459                  | 43,547               | 116,041     | 889,191   |
|               | Dec-15 | Mallards    | 6,103                              | 1,287       | 59,153                         | 17,784    | 107,474                | 109,493 | 13,682      | 5,814                          | 11,408                  | 9,242                | 5,837       | 347,277   |
|               |        | Total Ducks | 98,739                             | 25,214      | 106,887                        | 100,928   | 223,106                | 221,060 | 65,282      | 40,127                         | 21,975                  | 28,436               | 16,697      | 948,451   |
|               | MWS-16 | Mallards    | 31,506                             | 13,806      | 84,035                         | 14,558    | 53,900                 | 97,829  | 106,172     | 20,482                         | 60,454                  | -                    | 170,364     | 653,106   |
|               |        | Total Ducks | 55,172                             | 32,204      | 125,780                        | 37,662    | 91,665                 | 164,831 | 155,016     | 28,744                         | 74,250                  | 3,943                | 226,832     | 996,099   |
|               | Jan-16 | Mallards    | 22,606                             | 9,068       | 59,169                         | 22,800    | 80,590                 | 135,110 | -           | 116,169                        | -                       | 74,942               | 96,330      | 616,784   |
|               |        | Total Ducks | 94,269                             | 21,294      | 75,702                         | 33,212    | 105,643                | 184,233 | -           | 291,312                        | -                       | 74,942               | 111,648     | 992,255   |

Table 1, continued. Waterfowl abundance estimates in Arkansas during the late November (Nov), mid-December (Dec), early-January Midwinter Survey (MWS) and late-January (Jan) aerial waterfowl survey periods, 2009-2021, in the Mississippi Alluvial Valley (MAV) using stratified random sampling of transects.

|               |        |             | Survey Zone         |             |                |           |               |         |             |               |              |           |             |           |
|---------------|--------|-------------|---------------------|-------------|----------------|-----------|---------------|---------|-------------|---------------|--------------|-----------|-------------|-----------|
|               |        |             | Bayou Bartholomew - |             | Bayou Meto -   |           | Black - Upper |         |             | Lower White - | Little River | Lower St. |             |           |
|               |        |             | Bayou Boeuf         | Bayou Macon | Lower Arkansas | Big Creek | White         | Cache   | L' Anguille | Bayou Des Arc | Ditches      | Francis   | Lower White | MAV Total |
| Survey Period | Nov-15 | Mallards    | 3,599               | 43,200      | 17,915         | 19,253    | 15,382        | 46,418  | 7,625       | 15,597        | 9,093        | 40,889    | 42,941      | 261,912   |
|               |        | Total Ducks | 203,640             | 120,492     | 126,942        | 25,333    | 49,581        | 149,017 | 18,051      | 22,088        | 14,459       | 43,547    | 116,041     | 889,191   |
|               | Dec-15 | Mallards    | 6,103               | 1,287       | 59,153         | 17,784    | 107,474       | 109,493 | 13,682      | 5,814         | 11,408       | 9,242     | 5,837       | 347,277   |
|               |        | Total Ducks | 98,739              | 25,214      | 106,887        | 100,928   | 223,106       | 221,060 | 65,282      | 40,127        | 21,975       | 28,436    | 16,697      | 948,451   |
|               | MWS-16 | Mallards    | 31,506              | 13,806      | 84,035         | 14,558    | 53,900        | 97,829  | 106,172     | 20,482        | 60,454       | -         | 170,364     | 653,106   |
|               |        | Total Ducks | 55,172              | 32,204      | 125,780        | 37,662    | 91,665        | 164,831 | 155,016     | 28,744        | 74,250       | 3,943     | 226,832     | 996,099   |
|               | Jan-16 | Mallards    | 22,606              | 9,068       | 59,169         | 22,800    | 80,590        | 135,110 | -           | 116,169       | -            | 74,942    | 96,330      | 616,784   |
|               |        | Total Ducks | 94,269              | 21,294      | 75,702         | 33,212    | 105,643       | 184,233 | -           | 291,312       | -            | 74,942    | 111,648     | 992,255   |
|               | Nov-16 | Mallards    | 0                   | 0           | 26,781         | 21,094    | 0             | 1,792   | 3,007       | 285           | 16,572       | 0         | 12,381      | 81,912    |
|               |        | Total Ducks | 5,983               | 17,179      | 71,612         | 57,213    | 1,167         | 24,772  | 29,140      | 1,064         | 33,788       | 9,724     | 17,919      | 269,561   |
|               | Dec-16 | Mallards    | 15,104              | 475         | 150,591        | 31,456    | 23,246        | 91,324  | 19,088      | 8,160         | 20,241       | 20,767    | 64,914      | 445,364   |
|               |        | Total Ducks | 72,010              | 8,361       | 207,710        | 43,213    | 26,332        | 115,977 | 30,448      | 43,642        | 30,147       | 86,977    | 85,357      | 750,174   |
|               | MWS-17 | Mallards    | 72,405              | 40,448      | 219,106        | 22,908    | 14,102        | 128,174 | 20,651      | 12,460        | 8,873        | 41,202    | 70,677      | 651,004   |
|               |        | Total Ducks | 95,012              | 57,394      | 250,439        | 26,358    | 38,389        | 236,142 | 36,784      | 13,479        | 9,892        | 75,996    | 75,677      | 915,562   |
|               | Jan-17 | Mallards    | 7,154               | 15,135      | 146,710        | 20,187    | 41,860        | 159,212 | 47,507      | 19,013        | 8,116        | 31,646    | 63,039      | 559,579   |
|               |        | Total Ducks | 73,706              | 66,649      | 225,301        | 28,396    | 87,546        | 277,917 | 85,046      | 57,463        | 10,021       | 51,226    | 91,663      | 1,054,934 |
|               | Dec-17 | Mallards    | 4,921               | 3,151       | 116,026        | 19,729    | 84,718        | 38,466  | 26,874      | 2,400         | 26,662       | 100,522   | 9,508       | 432,977   |
|               |        | Total Ducks | 28,720              | 12,448      | 192,672        | 24,770    | 158,347       | 70,974  | 64,906      | 39,102        | 37,663       | 139,882   | 21,915      | 791,399   |
|               | MWS-18 | Mallards    | 2,458               | 34,577      | 390,205        | 92,504    | 40,402        | 132,049 | 35,330      | 1,402         | 12,274       | 54,505    | 153,625     | 949,331   |
|               |        | Total Ducks | 3,027               | 62,533      | 415,037        | 110,084   | 44,660        | 140,405 | 58,871      | 3,845         | 13,969       | 122,781   | 180,326     | 1,155,538 |
|               | Jan-18 | Mallards    | 3,276               | 10,690      | 104,937        | 116,012   | 8,117         | 21,688  | 11,050      | 555           | 36           | 70,030    | 63,378      | 409,769   |
|               |        | Total Ducks | 42,652              | 35,963      | 118,023        | 116,275   | 10,768        | 22,626  | 17,671      | 2,313         | 39           | 143,833   | 69,635      | 579,794   |
|               | Nov-18 | Mallards    | 251                 | 476         | 66,867         | 7,222     | 91,284        | 110,677 | 43,214      | 1,572         | 40,305       | -         | 2,226       | 364,094   |
|               |        | Total Ducks | 57,431              | 17,075      | 131,319        | 11,649    | 214,432       | 265,268 | 73,438      | 3,900         | 57,849       | 2,040     | 2,997       | 837,398   |
|               | Dec-18 | Mallards    | 2,770               | 7,210       | 118,723        | 124,685   | 33,242        | 145,660 | 84,416      | 9,825         | 31,723       | 45,074    | 83,800      | 687,126   |
|               |        | Total Ducks | 37,533              | 59,037      | 202,869        | 147,520   | 48,481        | 185,811 | 236,571     | 18,709        | 43,519       | 110,004   | 91,944      | 1,181,998 |
|               | MWS-19 | Mallards    | 50,569              | 7,541       | 80,381         | 22,208    | 81,122        | 85,902  | 38,201      | 16,263        | 13,588       | 119,119   | 40,885      | 555,779   |
|               |        | Total Ducks | 123,101             | 28,889      | 127,772        | 28,331    | 168,597       | 137,596 | 76,985      | 24,204        | 50,781       | 211,288   | 77,009      | 1,054,553 |
|               | Jan-19 | Mallards    | 3,592               | 12,942      | 91,603         | 21,192    | 53,730        | 132,098 | 49,299      | 19,298        | 3,937        | 50,190    | 117,601     | 555,481   |
|               |        | Total Ducks | 35,277              | 48,923      | 125,488        | 65,460    | 94,400        | 247,074 | 99,281      | 45,922        | 4,810        | 63,203    | 134,519     | 964,356   |
|               | Nov-19 | Mallards    | 4,095               | 2,656       | 18,897         | 10,217    | 12,387        | 102,483 | 28,188      | 8,764         | 40,098       | 1,068     | 13,419      | 242,272   |
|               |        | Total Ducks | 27,470              | 33,184      | 56,514         | 16,489    | 47,286        | 198,832 | 102,188     | 28,529        | 52,422       | 90,384    | 28,433      | 681,730   |
|               | Dec-19 | Mallards    | 3,141               | 1,410       | 60,346         | 24,176    | 73,688        | 103,543 | 19,682      | 9,443         | 70,195       | 60,545    | 20,914      | 447,083   |
|               |        | Total Ducks | 52,686              | 10,283      | 129,781        | 28,373    | 166,850       | 185,574 | 49,036      | 39,966        | 74,342       | 124,603   | 37,162      | 898,656   |
|               | MWS-20 | Mallards    | 40,730              | 7,032       | 49,576         | 10,209    | 56,228        | 76,464  | 57,146      | 40,810        | 11,322       | 333,707   | 46,736      | 729,960   |
|               |        | Total Ducks | 179,674             | 26,737      | 94,990         | 35,729    | 95,958        | 133,810 | 148,207     | 54,851        | 12,291       | 611,108   | 78,730      | 1,472,083 |
|               | Jan-20 | Mallards    | 4,225               | 7,189       | 96,105         | 26,071    | 130,185       | 47,918  | 28,615      | 3,329         | 26,127       | 116,120   | 61,002      | 546,885   |
|               |        | Total Ducks | 45,122              | 65,224      | 198,473        | 41,544    | 215,178       | 75,474  | 75,084      | 18,046        | 34,296       | 190,198   | 85,771      | 1,044,410 |
|               | Dec-20 | Mallards    | 17,899              | 86,795      | 13,611         | 8,565     | 3,869         | 38,361  | 20,225      | 9,044         | 7,128        | 36,802    | 12,753      | 255,052   |
|               |        | Total Ducks | 57,991              | 169,264     | 46,643         | 23,418    | 18,710        | 83,276  | 43,232      | 14,636        | 33,444       | 85,922    | 56,441      | 632,977   |
|               | MWS-21 | Mallards    | 36,123              | 37,667      | 93,865         | 10,850    | 63,320        | 42,077  | 3,288       | 2,028         | 70,340       | 31,220    | 67,535      | 458,314   |
|               |        | Total Ducks | 62,282              | 57,556      | 212,836        | 25,105    | 128,287       | 149,534 | 11,855      | 5,967         | 122,325      | 92,015    | 210,449     | 1,078,209 |
|               | Jan-21 | Mallards    | 8,591               | 24,694      | 82,346         | 11,291    | 23,970        | 53,046  | 38,576      | 15,881        | 18,335       | 94,037    | 157,153     | 527,920   |
|               |        | Total Ducks | 30,437              | 75,822      | 242,914        | 41,630    | 37,284        | 146,785 | 108,042     | 30,232        | 44,659       | 178,998   | 271,348     | 1,208,147 |
|               | Dec-21 | Mallards    | 6,097               | 3,141       | 38,795         | 1,977     | 24,843        | 85,582  | 13,647      | 1,544         | 12,401       | 40,941    | 97,096      | 326,064   |
|               |        | Total Ducks | 25,141              | 9,219       | 79,722         | 7,929     | 54,457        | 144,733 | 18,237      | 14,368        | 30,937       | 74,452    | 107,261     | 566,456   |

Table 2. Waterfowl abundance estimates in western Arkansas during the late November (Nov), mid-December (Dec), early-January Midwinter Survey (MWS) and late-January (Jan) aerial waterfowl survey periods, 2009-2021. Beginning in Jan. 2013, surveys in the Arkansas River Valley (ARV) were conducted using stratified random sampling of transects, while past ARV surveys and surveys in southwest Arkansas were conducted using “cruise” surveys.

|               |        |             | Survey Zone           |        |                           |                 |            |            |            |                          |                           |                             |        |
|---------------|--------|-------------|-----------------------|--------|---------------------------|-----------------|------------|------------|------------|--------------------------|---------------------------|-----------------------------|--------|
|               |        |             | Bigelow - Lake Conway | Cadron | East Dardanelle Reservoir | Fourche La Fave | Frog Bayou | Holla Bend | Petit Jean | Pt. Remove - Plumerville | West Dardanelle Reservoir | Arkansas River Valley Total |        |
| Survey Period | Nov-09 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 13,731                      | 5,480  |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 31,416                      | 19,140 |
|               | Dec-09 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 18,580                      | 19,230 |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 31,304                      | 31,820 |
|               | MWS-10 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 58,815                      | 34,590 |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 81,685                      | 36,060 |
|               | Jan-10 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 14,359                      | 19,840 |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 20,336                      | 27,705 |
|               | Nov-10 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 96                          | 14,010 |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 5,966                       | 30,300 |
|               | Dec-10 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 25,064                      | 2,390  |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 28,054                      | 21,106 |
|               | MWS-11 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 26,318                      | 15,027 |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 40,470                      | 21,267 |
|               | Jan-11 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 41,850                      | -      |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 60,635                      | -      |
|               | Nov-11 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 12,225                      | -      |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 19,870                      | -      |
|               | Dec-11 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 21,389                      | -      |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 40,919                      | -      |
|               | MWS-12 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 7,264                       | -      |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 13,339                      | -      |
|               | Jan-12 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 13,900                      | -      |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 21,000                      | -      |
|               | Nov-12 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 1,182                       | 13,090 |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 7,732                       | 21,935 |
|               | Dec-12 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 13,975                      | 10,245 |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 22,417                      | 17,105 |
|               | MWS-13 | Mallards    |                       |        |                           |                 |            |            |            |                          |                           | 16,893                      | 8,165  |
|               |        | Total Ducks |                       |        |                           |                 |            |            |            |                          |                           | 26,058                      | 14,630 |
|               | Jan-13 | Mallards    | -                     | 408    | 10,000                    | 372             | 1,837      | 630        | 627        | 1,843                    | 917                       | 16,634                      | -      |
|               |        | Total Ducks | -                     | 1,428  | 10,180                    | 372             | 1,971      | 990        | 902        | 3,687                    | 7,857                     | 28,011                      | -      |
|               | Nov-13 | Mallards    | 240                   | 187    | 4,660                     | 800             | 0          | 144        | 0          | 754                      | 253                       | 7,038                       | 4,455  |
|               |        | Total Ducks | 320                   | 187    | 14,320                    | 1,920           | 0          | 1,080      | 528        | 965                      | 3,307                     | 22,627                      | 19,145 |
|               | Dec-13 | Mallards    | 576                   | 245    | 5,472                     | 1,728           | 358        | 162        | 1,320      | 3,429                    | 2,176                     | 15,466                      | 10,130 |
|               |        | Total Ducks | 1,604                 | 2,713  | 8,672                     | 1,728           | 1,836      | 3,132      | 1,501      | 4,329                    | 3,941                     | 29,456                      | 29,070 |
|               | MWS-14 | Mallards    | 11,767                | 816    | 2,898                     | 4,800           | -          | 2,160      | 715        | 13,703                   | 3,449                     | 40,306                      | 18,385 |
|               |        | Total Ducks | 14,441                | 816    | 8,711                     | 5,124           | -          | 2,934      | 957        | 22,177                   | 6,087                     | 61,247                      | 35,875 |
|               | Nov-14 | Mallards    | 926                   | 7,140  | 12,114                    | 704             | 924        | 4,518      | 10,428     | 7,125                    | 392                       | 44,271                      | 15,890 |
|               |        | Total Ducks | 5,040                 | 10,540 | 45,485                    | 4,256           | 3,248      | 4,518      | 19,932     | 12,039                   | 624                       | 105,682                     | 29,790 |
|               | Dec-14 | Mallards    | 720                   | 224    | 1,028                     | 640             | 373        | 3,006      | 2,541      | 1,343                    | 299                       | 10,174                      | 21,200 |
|               |        | Total Ducks | 1,242                 | 530    | 33,805                    | 1,296           | 373        | 4,194      | 4,059      | 6,991                    | 299                       | 52,789                      | 29,400 |
|               | MWS-15 | Mallards    | 3,929                 | 143    | 5,813                     | 221             | -          | 11,138     | 0          | 2,107                    | 3,531                     | 26,882                      | 19,245 |
|               |        | Total Ducks | 10,594                | 755    | 18,649                    | 221             | -          | 13,455     | 224        | 2,107                    | 9,871                     | 55,876                      | 28,695 |
|               | Nov-15 | Mallards    | 270                   | -      | 1,867                     | -               | 149        | 2,430      | 561        | 4,785                    | 64                        | 10,126                      | 21,580 |
|               |        | Total Ducks | 270                   | 449    | 2,898                     | -               | 1,170      | 14,760     | 726        | 7,042                    | 64                        | 27,379                      | 37,060 |
|               | Dec-15 | Mallards    | 1,440                 | 340    | 320                       | 160             | 140        | 563        | 165        | 2,864                    | 1,027                     | 7,019                       | 11,425 |
|               |        | Total Ducks | 4,140                 | 374    | 3,140                     | 992             | 140        | 7,088      | 165        | 6,913                    | 3,274                     | 26,226                      | 17,950 |
|               | MWS-16 | Mallards    | 411                   | 775    | 352                       | 496             | 14,000     | 3,042      | 726        | 2,544                    | 6,070                     | 28,416                      | 10,310 |
|               |        | Total Ducks | 617                   | 775    | 6,752                     | 896             | 17,562     | 6,102      | 990        | 3,808                    | 15,019                    | 52,521                      | 16,715 |
|               | Jan-16 | Mallards    | 634                   | 918    | 2,743                     | 576             | 373        | 1,548      | 14,388     | 8,479                    | 4,622                     | 34,281                      | 14,735 |
|               |        | Total Ducks | 634                   | 918    | 3,817                     | 1,536           | 1,966      | 2,088      | 18,777     | 11,815                   | 5,478                     | 47,029                      | 19,565 |

Table 2, continued. Waterfowl abundance estimates in western Arkansas during the late November (Nov), mid-December (Dec), early-January Midwinter Survey (MWS) and late-January (Jan) aerial waterfowl survey periods, 2009-2021. Beginning in Jan. 2013, surveys in the Arkansas River Valley (ARV) were conducted using stratified random sampling of transects, while past ARV surveys and surveys in southwest Arkansas were conducted using “cruise” surveys. The Bigelow-Lake Conway and Cadron strata were not flown beginning December, 2019.

|               |        |             | Survey Zone              |        |                              |                    |            |            |            |                             |                              |                                |                             |
|---------------|--------|-------------|--------------------------|--------|------------------------------|--------------------|------------|------------|------------|-----------------------------|------------------------------|--------------------------------|-----------------------------|
|               |        |             | Bigelow - Lake<br>Conway | Cadron | East Dardanelle<br>Reservoir | Fourche La<br>Fave | Frog Bayou | Holla Bend | Petit Jean | Pt. Remove -<br>Plumerville | West Dardanelle<br>Reservoir | Arkansas River<br>Valley Total | Southwest<br>Arkansas Total |
| Survey Period | Nov-16 | Mallards    | -                        | -      | 818                          | -                  | 0          | -          | -          | -                           | 99                           | 917                            | 5,165                       |
|               |        | Total Ducks | -                        | -      | 6,530                        | -                  | 814        | -          | -          | -                           | 100                          | 7,444                          | 14,690                      |
|               | Dec-16 | Mallards    | 112                      | -      | -                            | 739                | 187        | 2,612      | 296        | 234                         | 8,186                        | 12,364                         | 34,946                      |
|               |        | Total Ducks | 333                      | -      | 3,165                        | 1,016              | 988        | 3,248      | 550        | 1,788                       | 10,192                       | 21,278                         | 39,360                      |
|               | MWS-17 | Mallards    | 24                       | 1,538  | 180                          | 831                | 242        | 448        | 5,050      | 1,808                       | 2,333                        | 12,454                         | 19,386                      |
|               |        | Total Ducks | 325                      | 2,137  | 453                          | 12,788             | 2,167      | 547        | 5,499      | 4,461                       | 14,900                       | 43,277                         | 31,679                      |
|               | Jan-17 | Mallards    | 17                       | 627    | 16,432                       | 3,812              | 1,019      | 5,394      | 1,561      | 14,818                      | 4,768                        | 48,448                         | 13,682                      |
|               |        | Total Ducks | 17                       | 1,647  | 17,810                       | 11,308             | 2,595      | 5,638      | 1,825      | 14,836                      | 4,917                        | 60,593                         | 26,594                      |
|               | Dec-17 | Mallards    | -                        | -      | 821                          | -                  | 0          | 1,184      | -          | -                           | 2,129                        | 4,134                          | 15,487                      |
|               |        | Total Ducks | -                        | -      | 2,558                        | -                  | 2,972      | 3,654      | -          | -                           | 4,264                        | 13,448                         | 34,822                      |
|               | MWS-18 | Mallards    | 0                        | 0      | 10,862                       | 1,013              | 4,784      | 22,254     | 0          | 5,269                       | 6,711                        | 50,893                         | 18,412                      |
|               |        | Total Ducks | 510                      | 0      | 13,785                       | 2,114              | 5,880      | 36,695     | 0          | 13,843                      | 7,553                        | 80,380                         | 38,114                      |
|               | Jan-18 | Mallards    | 2,080                    | 3,144  | 11,881                       | 135                | 1,115      | 141,074    | 845        | 3,361                       | 5,214                        | 168,849                        | 10,849                      |
|               |        | Total Ducks | 3,420                    | 4,489  | 20,281                       | 227                | 3,826      | 174,542    | 3,150      | 3,313                       | 5,381                        | 218,629                        | 32,928                      |
|               | Nov-18 | Mallards    | -                        | -      | 273                          | 2,956              | 3,617      | 198        | 4,733      | 7,074                       | 429                          | 19,280                         | 9,721                       |
|               |        | Total Ducks | -                        | -      | 5,878                        | 3,319              | 3,895      | 253        | 8,867      | 9,956                       | 502                          | 32,670                         | 26,969                      |
|               | Dec-18 | Mallards    | 235                      | 326    | 2,440                        | 73                 | 179        | 3,292      | 462        | 7,426                       | 605                          | 15,038                         | 9,241                       |
|               |        | Total Ducks | 240                      | 330    | 4,483                        | 73                 | 630        | 3,472      | 1,771      | 10,920                      | 605                          | 22,514                         | 35,236                      |
|               | MWS-19 | Mallards    | 58                       | 382    | 841                          | 120                | 389        | 89         | 2,413      | 9,527                       | 4,418                        | 18,237                         | 3,507                       |
|               |        | Total Ducks | 58                       | 748    | 4,417                        | 192                | 2,446      | 100        | 3,875      | 23,206                      | 4,582                        | 39,620                         | 30,973                      |
|               | Jan-19 | Mallards    | 1,628                    | -      | 1,603                        | 169                | 728        | 607        | 2,234      | 1,928                       | 7,900                        | 16,797                         | 5,978                       |
|               |        | Total Ducks | 5,295                    | -      | 2,252                        | 2,762              | 869        | 785        | 2,381      | 2,488                       | 10,513                       | 27,347                         | 25,540                      |
|               | Nov-19 | Mallards    | 1,096                    | -      | 1,388                        | 243                | 4,235      | 550        | 592        | 744                         | 608                          | 9,456                          | 8,016                       |
|               |        | Total Ducks | 1,081                    | -      | 2,067                        | 1,242              | 20,195     | 3,054      | 1,534      | 5,742                       | 1,754                        | 36,670                         | 17,193                      |
|               | Dec-19 | Mallards    | -                        | -      | 461                          | 363                | 1,418      | 1,171      | 2,121      | 2,133                       | 3,377                        | 11,044                         | -                           |
|               |        | Total Ducks | -                        | -      | 637                          | 672                | 1,595      | 2,286      | 3,146      | 4,160                       | 8,194                        | 20,691                         | -                           |
|               | MWS-20 | Mallards    | -                        | -      | 7,995                        | 617                | 277        | 6,274      | 655        | 4,907                       | 196                          | 20,920                         | 2,237                       |
|               |        | Total Ducks | -                        | -      | 10,702                       | 917                | 2,711      | 9,008      | 2,641      | 7,409                       | 959                          | 34,346                         | 26,955                      |
|               | Jan-20 | Mallards    | -                        | -      | 561                          | 460                | 4,227      | 1,339      | 795        | 1,465                       | 4,153                        | 13,000                         | 1,960                       |
|               |        | Total Ducks | -                        | -      | 4,792                        | 2,567              | 5,108      | 1,633      | 1,397      | 3,604                       | 6,254                        | 25,355                         | 18,480                      |
|               | Dec-20 | Mallards    | -                        | -      | 1,452                        | 294                | 561        | 131        | 1,123      | 2,538                       | 1,313                        | 7,416                          | 12,367                      |
|               |        | Total Ducks | -                        | -      | 4,606                        | 471                | 2,276      | 205        | 2,059      | 4,722                       | 2,616                        | 16,958                         | 45,832                      |
|               | MWS-21 | Mallards    | -                        | -      | 0                            | 1,594              | 104        | 53         | 707        | 1,592                       | 5,315                        | 9,366                          | 4,597                       |
|               |        | Total Ducks | -                        | -      | 1,647                        | 1,864              | 645        | 337        | 828        | 5,019                       | 11,017                       | 21,357                         | 9,104                       |
|               | Jan-21 | Mallards    | -                        | -      | 729                          | 292                | 2,451      | 2,381      | 330        | 47,577                      | 1,383                        | 55,141                         | 5,846                       |
|               |        | Total Ducks | -                        | -      | 1,741                        | 962                | 3,125      | 2,707      | 544        | 55,122                      | 5,212                        | 69,413                         | 49,122                      |
|               | Dec-21 | Mallards    | -                        | -      | 1,422                        | 250                | 326        | 1,760      | 131        | 797                         | 2,033                        | 6,719                          | 10,019                      |
|               |        | Total Ducks | -                        | -      | 6,058                        | 919                | 384        | 1,739      | 1,137      | 806                         | 2,391                        | 13,434                         | 54,031                      |

Figure 1. Duck abundance estimates in the Mississippi Alluvial Valley (Delta) of Arkansas during the late November (Nov), mid-December (Dec), early-January Midwinter Waterfowl Survey (MWS) and late-January (Jan) aerial waterfowl survey periods, 2009-2021.

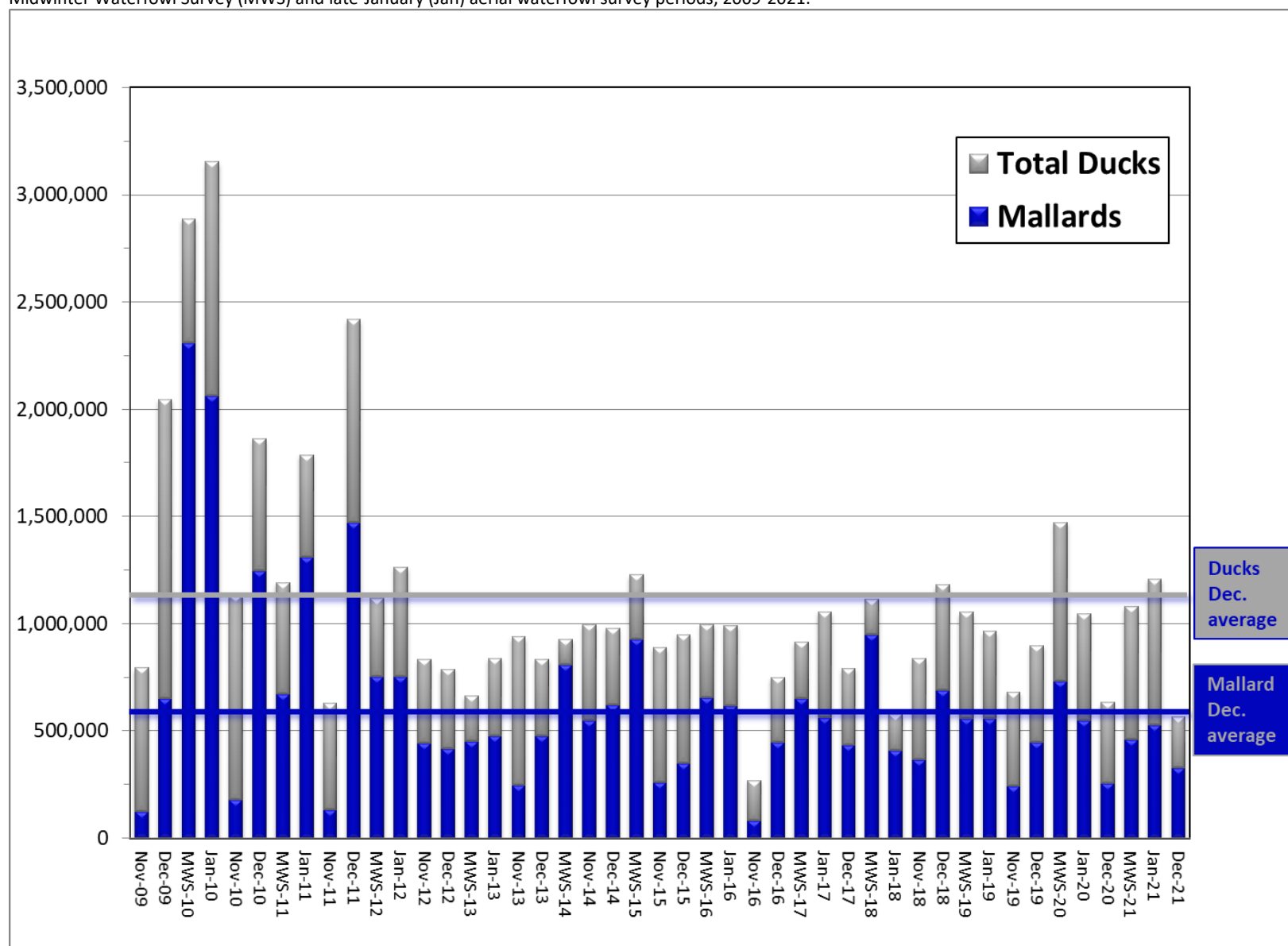


Figure 2. Duck abundance estimates in the Arkansas River valley of Arkansas during the late November (Nov), mid-December (Dec), early-January Midwinter Waterfowl Survey (MWS) and late-January (Jan) aerial waterfowl survey periods, 2009-2021.

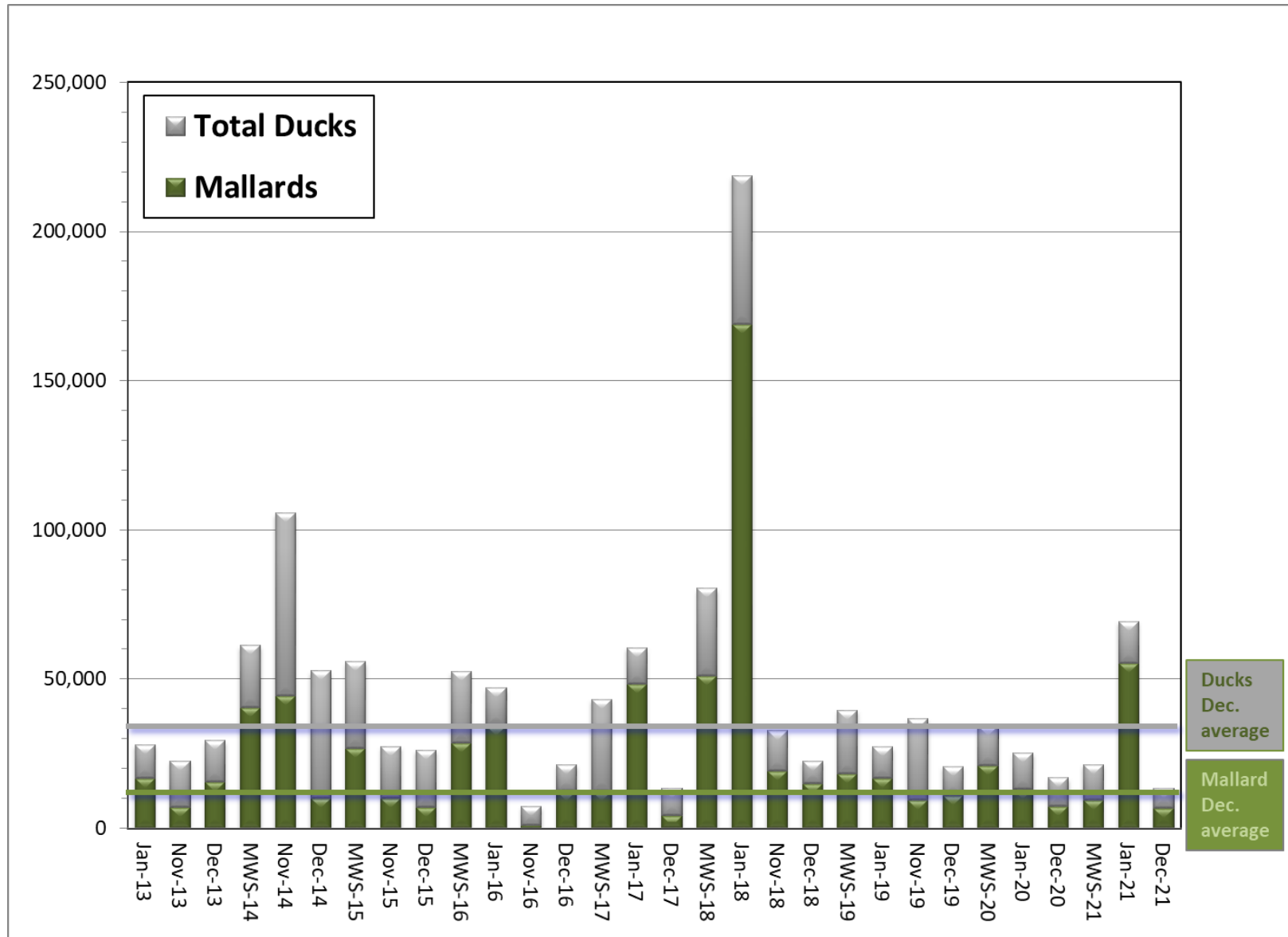


Figure 3. Duck distribution in the Mississippi Alluvial Valley of Arkansas during the December 2021 aerial waterfowl survey period.

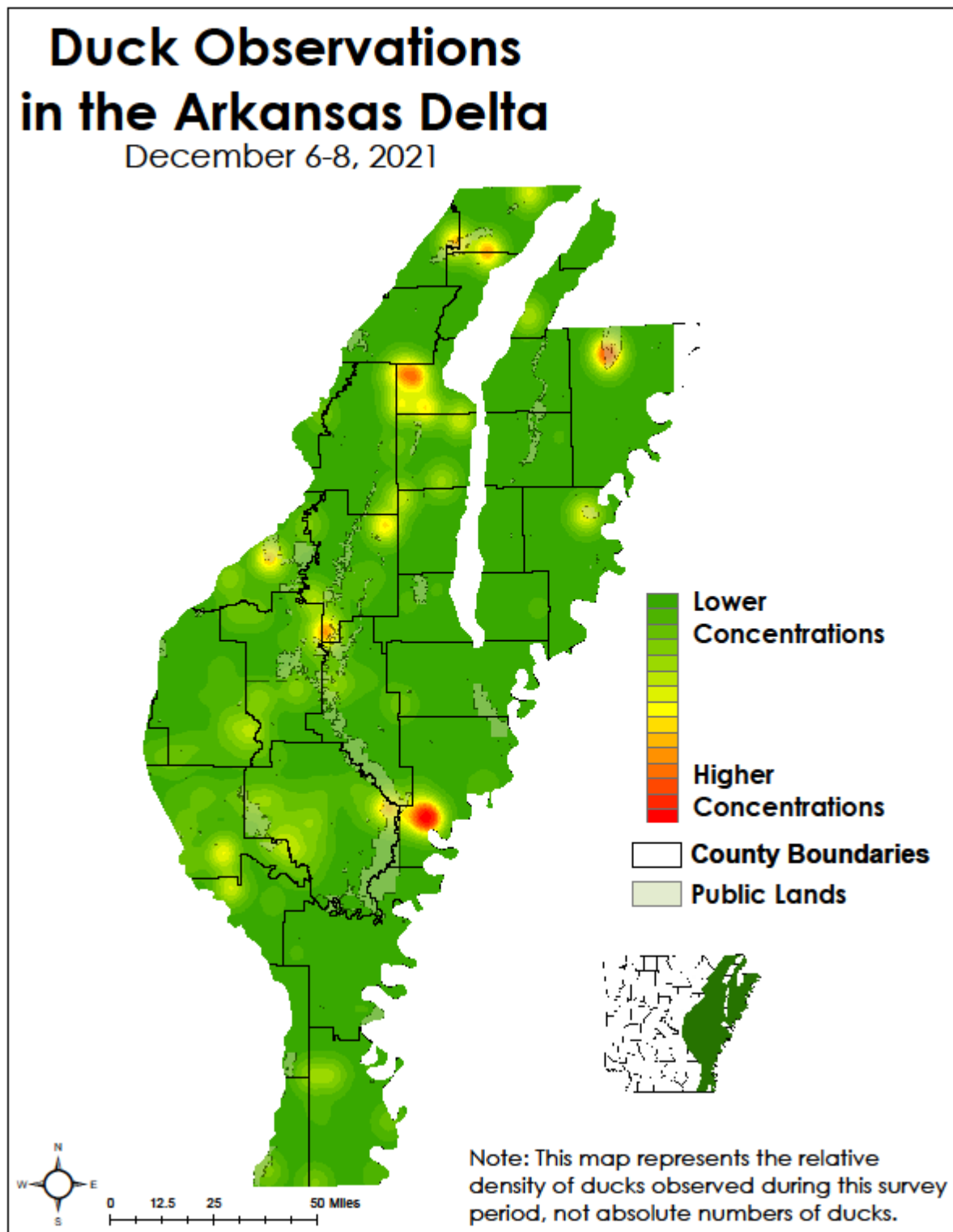




Figure 4. Mallard distribution in the Mississippi Alluvial Valley of Arkansas during the December 2021 aerial waterfowl survey period.

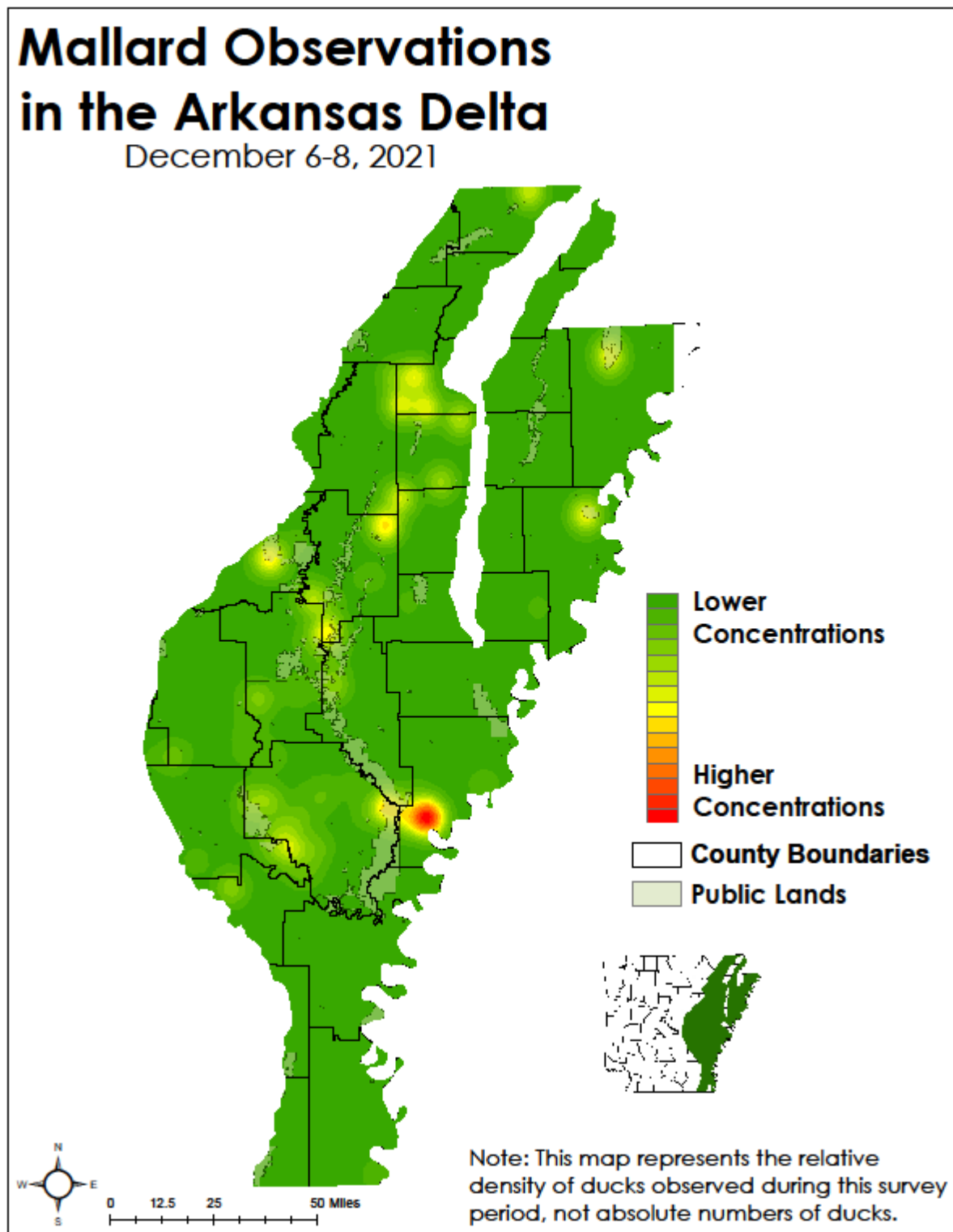


Figure 5. Duck distribution in the Arkansas River Valley (ARV) of Arkansas during the December 2021 waterfowl survey period.

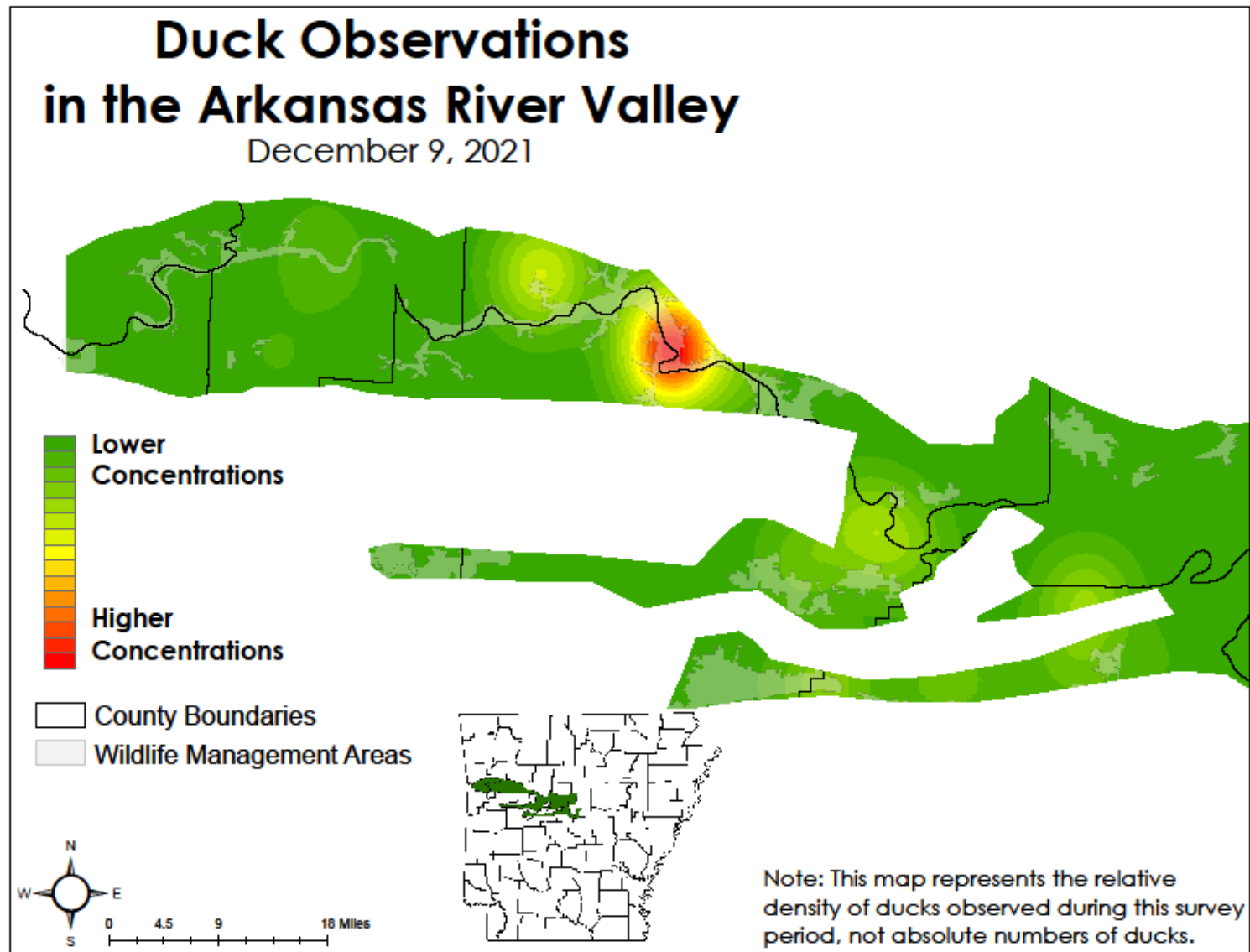
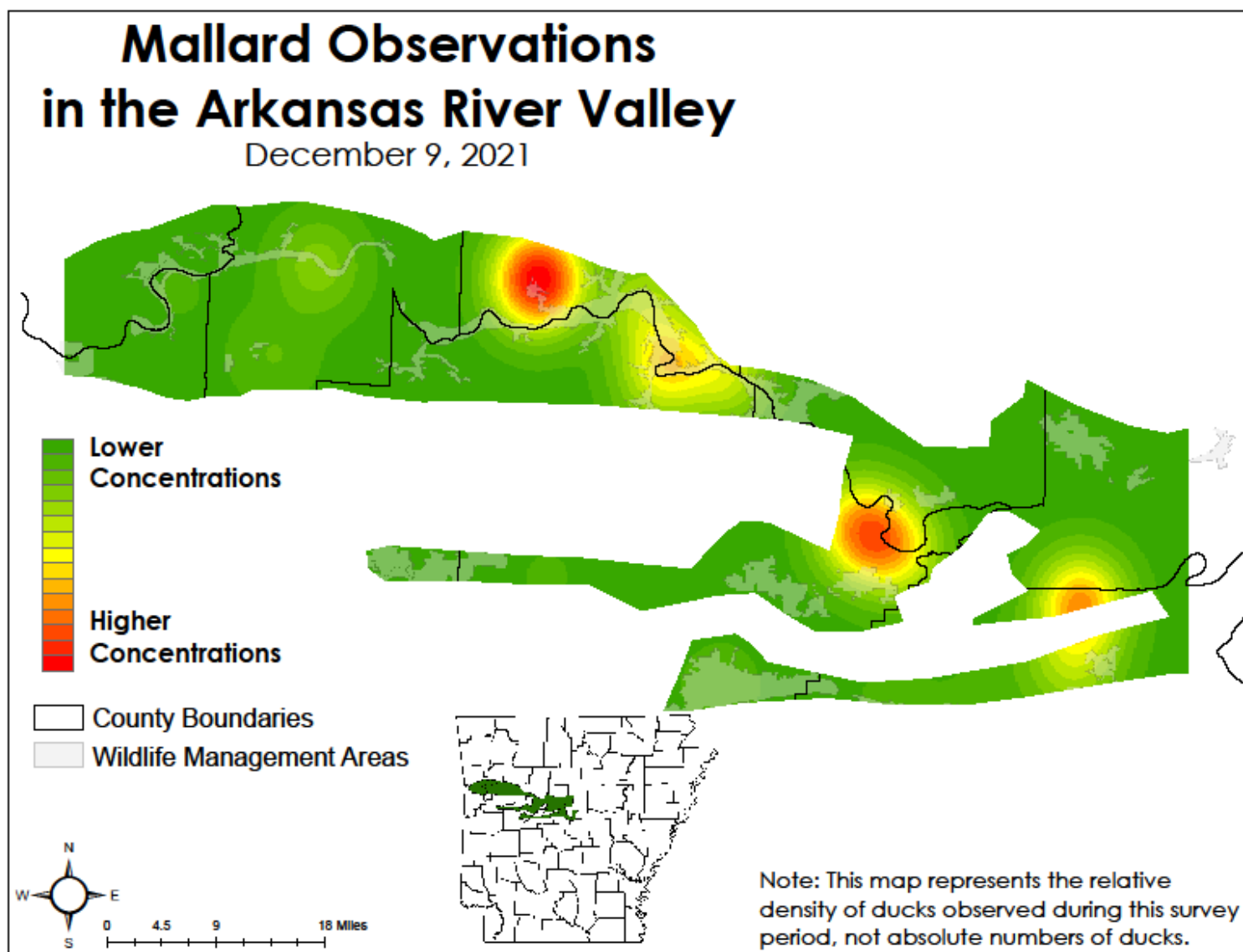


Figure 6. Mallard distribution in the Arkansas River Valley (ARV) of Arkansas during the December 2021 waterfowl survey period.



## **Survey Design Background**

The Mississippi Alluvial Valley is an area of continental significance for migrating and wintering waterfowl, as outlined in the North American Waterfowl Management Plan, and the single most important region for wintering mallards. Habitats found in western Arkansas, including the Arkansas River Valley and southwest Arkansas, such as the Red River and Sulphur River floodplains, provide additional critical habitat for migrating and wintering waterfowl. Biologists conduct regular waterfowl surveys in these regions by aircraft up to four times each wintering period.

Winter waterfowl surveys, including the Midwinter Waterfowl Survey, have been conducted across much of the United States since 1935. Many different counting techniques have been used, and recently AGFC and partners have conducted surveys in the MAV using stratified random sampling of aerial fixed width (250m) strips, or transects, that have the advantages of extensive coverage (i.e., no area is excluded from the sample), increased accuracy by counting on fixed strips rather than traditional “cruise” surveys only counting waterfowl on large concentration areas, and availability of measures of sampling error.

Beginning in 2011 in the MAV, survey strata – or sampling zones – follow watershed boundaries (Figure 4). Watersheds in this case are simply land areas that are occupied by a drainage system consisting of a portion of a surface stream and all the tributary surface streams feeding it. For example, the Cache River strata includes lands surrounding and tributaries flowing into the Cache River from the Missouri border on the north to the Cache River’s junction with the White River on the south. At the root of this sampling design is the idea that habitat within these zones will share common weather and flooding patterns and, knowing that ducks are keyed in on such patterns, duck distribution will vary among watersheds. This is not a concept foreign to those who follow ducks, particularly duck hunters, as they frequently discuss habitat and duck numbers in terms of conditions in the “Cache River bottoms,” for instance. Systematically conducting aerial waterfowl surveys using this design will allow for more efficient allocation of sampling effort and provide precise estimates of waterfowl abundance in the MAV. Such a design offers an opportunity to track changes in abundance in response to changes in land use, flooding patterns or weather conditions, for example.

Before each survey period, transects to be flown are randomly selected within each strata. Biologists spend many hours in the air flying each of these transects – totaling over 3,500 miles each survey – recording all waterfowl observations using specialized computer software that collects location information in flight. Biologists also collect habitat information for each duck observation to track trends in habitat use. These data can then be used to generate population estimates for each strata and the entire MAV and develop visual representations of duck distribution (i.e., duck density maps).

Figure 6. Aerial waterfowl survey strata in the Mississippi Alluvial Valley (Delta) of Arkansas.

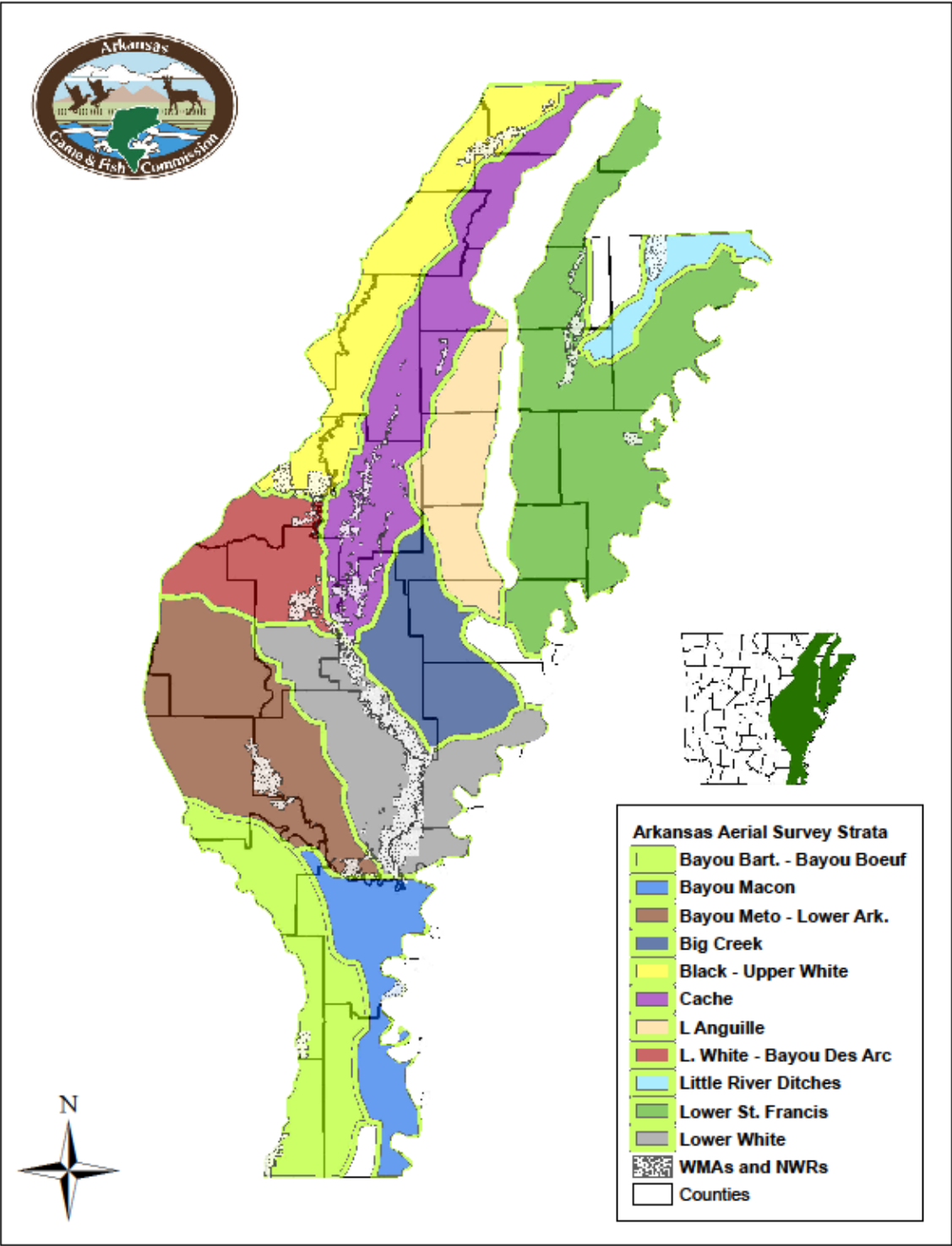


Figure 7. Aerial waterfowl survey strata in the Arkansas River valley (ARV) of western Arkansas.

